

determine the complete past and future of the universe with certainty. If so, then any event—winning the lottery, running into a long-lost friend on the other side of the world, being hit by a meteor—is not actually random, as we might be tempted to think of it, but merely the fully predetermined consequence of the way the universe was set up around a dozen billion years ago. This would mean that there is no randomness, that we are stuck in a player piano’s melody and things appear random only because we don’t know enough about them.

The mathematical rules of physics as we understand them are consistent with a deterministic universe, but they are also consistent with a nondeterministic universe in which randomness really does exist and, as some have put it, God “plays dice.” They are also consistent with a “many worlds” scenario in which every possible version of an event occurs, but in different universes that are inaccessible from each other. All these interpretations of the laws of physics are further complicated if we try to find a place for free will in the cosmos. The interpretation of mathematical physics that we accept depends not on our mathematical understanding but rather on our philosophical inclinations—any position is acceptable mathematically.

Whether or not the universe itself contains randomness, your laptop doesn’t—or at least it isn’t supposed to. Computers are meant to be our perfectly obedient servants and do only what we explicitly command them to do, exactly when and how we command them to do it. To ask a computer to run a video game, perform machine learning via a random forest, or administer a randomized experiment is to ask a supposedly deterministic machine to generate something nondeterministic: a random number. This is an impossible request.

Since a computer cannot deliver true randomness, we’ve designed algorithms that can deliver the next-best thing: *pseudorandomness*. Pseudorandom number generation algorithms are important for all the reasons that random numbers are important. Since true randomness is impossible on a computer (and may be impossible in the universe at large), pseudorandom number generation algorithms must be designed with great care so that their outputs resemble true randomness as closely as possible. The way we judge whether a pseudorandom number generation algorithm